Design of Quadricycle Vehicle and Mobility Service

TEAM 2 - VEHICLE DYNAMICS



Vehicle Specification	
Dry mass	450 kg
Gross weight	800 kg
Wheelbase	2.0 m
Track width	1.295 m
CoG height	0.53 m
Weight distribution	55/45 (F/R)
Design speed	90 km/h

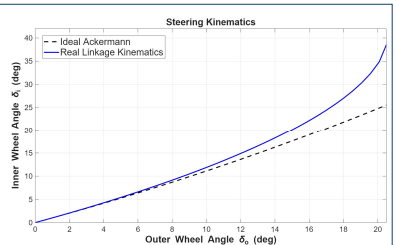
Project Scope

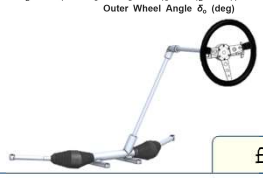
- ❑ Conceptual design of a quadricycle vehicle (L7e-CP Category).
- ❑ Alternative to rental scooters for urban/inter-urban trips (Cranfield).
- ❑ Minimise vehicle size and emissions.

- ❑ Design and analysis of suspensions, brakes, steering and tyres.
- ❑ Ride & handling behaviour (safety, payload sensitivity, comfort).
- ❑ Package integration and design with other groups (Powertrains & Chassis).

Steering

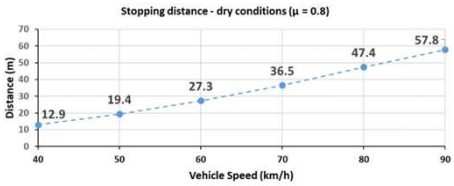
Mechanism	Rack & Pinion
Turning Circle (Kerb-To-Kerb)	Ø10.2 m
Assist Type	Non-assist
Steering Ratio	22.9:1
Steering effort	6.6 daN
Steering wheel turns (lock-to-lock)	3.5 turns
Steering column	Collapsible
System weight	8 kg





£ 300

Brakes

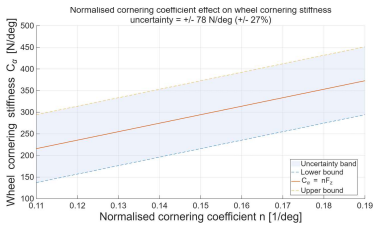


	Front disc	Rear drum
Force	3600 N	1490 N
Brake factor	1.0	2.5
Brake torque	490 Nm	200 Nm
Material	Cast iron	Cast iron

Pedal force	350 N
Max deceleration	5.8 m/s²
Brake split	70 / 30
Master cylinder	Dual Piston Floating (Ø20)

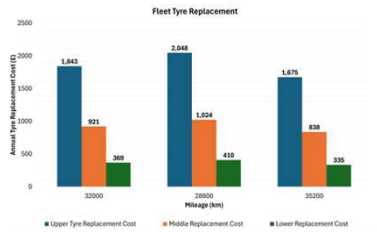
£ 500

Wheels & Tyres





Size	155/65 R13
Load index	73 (365 kg)
Speed index	T (190 km/h)
Mass	5 – 6 kg
Wet Grip	C
Efficiency	D





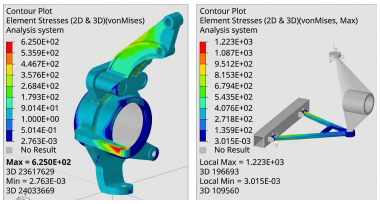
Wheel type	Steel
Size	13”x4.5J” ET36
Mass	6 – 7 kg
PCD	4 x 100 mm
Bore Φ	60 mm

£ 450

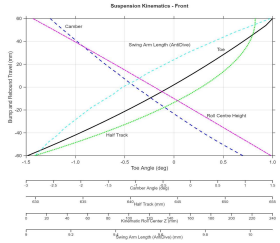
Suspension

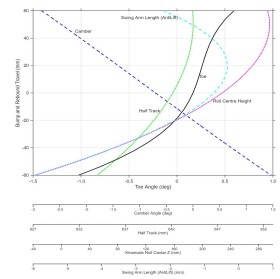
Front – McPherson	
Scrub radius	(-ve) 11 mm
Caster angle	9.9 deg
King Pin Inclination	17 deg
Roll Centre Height	137 mm
Static Camber	(-ve) 1 deg
Anti-Dive%	20%

Rear – McPherson	
Scrub radius	(+ve) 5 mm
Caster angle	3.8 deg
King Pin Inclination	11 deg
Roll Centre Height	114 mm
Static Camber	-(ve) 0.8 deg
Anti-Lift%	6%



£ 1800





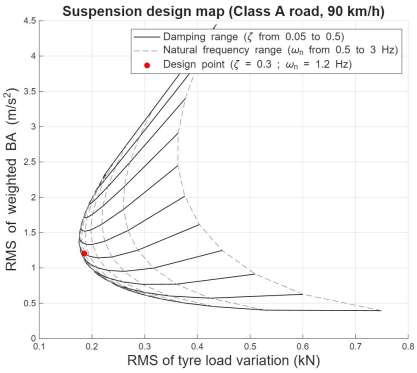


Front Knuckle Driven Axle



Rear Knuckle Non-driven Axle

Ride & Handling



Front Spring Stiffness	13100 N/m
Rear Spring Stiffness	14600 N/m
Front Damper ζ	0.3
Rear Damper ζ	0.2
Front ARB Stiffness	12300 N/m
Rear ARB Stiffness	8100 N/m
Roll Gradient @1.2g	5.1 Deg/g

Road class	Front BA RMS	Rear BA RMS
A	0.29	0.33
B	0.58	0.65
C	1.18	1.35

BA RMS (m/s²) @60 km/h

CarMaker test plan	
Braking stopping distance	ISO 21994
Split-μ braking	ISO 14512
Roll-over (Fishhook)	NHTSA
Steady-state cornering	ISO 4138
Slalom	ISO 3888-2

→ Payload sensitivity: 550kg / 800 kg



1. Stability & safety 2. Payload sensitivity 3. Ride dynamics

Conclusion

- ❑ Predictable handling (Understeer) characteristics exhibited in calculations and simulations (under steady-state conditions).
- ❑ Optimised package for minimal footprint under the given powertrain and chassis constraints.
- ❑ Comfort oriented ride (BA and SWS target met).
- ❑ Limited tyre and handling data restricted the ability to fully validate handling characteristics.

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